Hospital Patient Data Analysis with SQL

Analysis Report

Prepared by: Aisha Hamza | Date: 10–13 October 2026 | Tool: Microsoft SQL Server (SSMS) | Data period: January 2026

Executive Summary

This report presents an analysis of a hospital patient dataset using SQL Server. Twenty-five queries were written to answer questions about patient demographics, admissions, length of stay, medical conditions and doctor workload. The data holds 10 patients, 3 doctors and patients from 8 cities, covering admissions in January 2026.

The main findings are:

- **Admission type:** scheduled admissions stay 6.5 days on average, against 3.0 days for emergency admissions. Scheduled is the only type above 4 days.
- **Longest stays:** the three longest stays (8, 7 and 6 days) are all scheduled admissions for hypertension or diabetes, in patients aged 56 or older.
- **Conditions:** the 10 patients are split evenly across five conditions, with 2 patients each.
- **Doctors:** Dr. Ade treated the most patients (4) and has the longest average stay (5.0 days).
- **Age:** the average patient age is 44.6 years, with 4 Young, 4 Middle_age and 2 Senior patients.

1. Introduction and Objectives

Hospitals record who is admitted, why, for how long and under whose care. Analysing that data helps management plan beds, staffing and specialist coverage. This assignment used SQL to answer practical questions from a patient dataset. The objectives were to:

1. Find specific groups of patients by city, age, gender, condition and admission type.
2. Measure length of stay and see which admission types and doctors are linked to longer stays.
3. Count patients by medical condition and by age group.
4. Use date functions to see when patients were admitted.
5. Combine patient and doctor data to see how work is shared between doctors.

2. Data Description

The analysis uses two related tables in a SQL Server database called TWDP Assignment. The tables are linked by Doctor_ID, and every patient has a matching doctor. The data has no missing values. It is a practice dataset and does not contain real patient data.

Patients (10 rows)

Column	Description
Patient_ID	Unique identifier for each patient
Patient_Name	Patient's full name
Age	Patient's age in years
Gender	Female or Male
City	City where the patient lives
Condition	Malaria, Hypertension, Typhoid, Diabetes or Asthma
Admission_Type	Emergency or Scheduled
Doctor_ID	Links to the Doctors table
Admission_Date	Date of admission
Length_of_Stay	Number of days spent in hospital
Treatment_Status	Recovered or Recovering

Doctors (3 rows)

Column	Description
Doctor_ID	Unique identifier for each doctor
Doctor_Name	Doctor's name
Specialization	General Medicine, Internal Medicine or Family Medicine

3. Methodology

The CSV files were imported into SQL Server and checked with a `SELECT *` query to confirm they loaded correctly. The 25 queries were then written in SQL Server Management Studio and grouped by the SQL skill they use:

- **Filtering:** `WHERE`, `BETWEEN`, `AND` and `OR` (queries 1 to 5)
- **Sorting and unique values:** `ORDER BY`, `TOP` and `DISTINCT` (queries 6 and 7)
- **Aggregate functions:** `COUNT`, `SUM` and `GROUP BY` (queries 8 to 10)
- **String functions:** `UPPER`, `LOWER` and `CONCAT` (queries 11 to 13)
- **Date functions:** date comparison, `YEAR` and `DATENAME` (queries 14 and 15)
- **Grouping and filtering groups:** `GROUP BY` with `HAVING` (queries 16, 17 and 21)
- **Conditional logic:** `CASE` expressions (queries 18 to 20)
- **Joins:** `JOIN` on Doctor_ID, combined with aggregation (queries 22 to 25)

Averages were calculated with decimals (for example `SUM(Age) * 1.0 / COUNT(*)`), so results are not rounded down to whole numbers. In query 17, `AVG(Length_of_Stay * 1.0)` was used for the same reason, because AVG on whole numbers rounds down in SQL Server, which could wrongly exclude a group from a check such as "greater than 4". This does not change the output on this dataset, but it keeps the query correct on other data.

4. Analysis and Findings

4.1 Filtering patients

Three patients are from Lagos, which makes it the largest patient city. Three patients stayed longer than 5 days, and four are aged between 30 and 50. Two patients are female, from Lagos and admitted as emergencies. Four patients have diabetes or hypertension. All three stays longer than 5 days belong to patients with diabetes or hypertension, and all three are scheduled admissions. The fourth chronic patient stayed exactly 5 days.

4.2 Sorting and unique values

The three longest stays are P007 (8 days, hypertension, age 68), P004 (7 days, diabetes, age 61) and P009 (6 days, diabetes, age 56). The patients come from 8 cities: Abuja, Accra, Enugu, Ibadan, Kaduna, Kano, Lagos and Port Harcourt. Seven of the eight cities have only one patient each.

4.3 Aggregate summaries

There are 10 patients with an average age of 44.6 years. Each of the five conditions (Asthma, Diabetes, Hypertension, Malaria and Typhoid) has exactly 2 patients, so no single condition dominates.

4.4 String and date functions

Patient names were converted to uppercase and lowercase, and Patient_ID and Patient_Name were combined into one value (for example, "P001 Aisha Bello"). Five patients (P006 to P010) were admitted after 15 January 2026. P005, admitted on 15 January, is excluded because the filter uses "greater than". All 10 admissions fall in January 2026, roughly one new patient every two to three days.

4.5 Length of stay by admission type

Admission type	Patients	Average stay (days)
Emergency	6	3.0
Scheduled	4	6.5

Only scheduled admissions average more than 4 days. They stay more than twice as long as emergency admissions, so they use far more bed-days per patient. All five conditions have more than one patient, so query 21 returns all five.

4.6 Age groups and stay length

Age group	Patients	Definition
Young	4	Under 40
Middle_age	4	40 to 59
Senior	2	60 and above

Four patients had a short stay (3 days or less) and six had a long stay. The two seniors (P004, aged 61, and P007, aged 68) have the second and first longest stays, so age and length of stay move together in this dataset.

4.7 Doctors

Doctor	Specialization	Patients	Average stay (days)
Dr. Ade	General Medicine	4	5.0
Dr. Bello	Family Medicine	3	4.3
Dr. Musa	Internal Medicine	3	3.7

All 10 patients matched a doctor. Dr. Musa, the Internal Medicine specialist, treated 3 patients (P002, P005 and P008). Dr. Ade has the heaviest load, with the most patients and the longest average stay. The difference is partly down to case mix, since Dr. Ade also has the two oldest patients.

4.8 Additional observation

None of the 25 queries used the Treatment_Status column, but it shows a clear pattern: all 6 emergency patients are marked Recovered, and all 4 scheduled patients are still Recovering.

5. Key Findings

- The dataset has **10 patients**, **3 doctors** and patients from **8 cities**, with Lagos the largest (3 patients).
- The average age is **44.6 years**, and patients are split evenly across five conditions.
- **Scheduled admissions stay 6.5 days on average, against 3.0 days for emergencies.**
- The **three longest stays** are scheduled admissions for hypertension or diabetes in patients aged 56 or older.
- **Dr. Ade** treated the most patients (4) and has the longest average stay (5.0 days).
- Six of the 10 patients had a long stay (more than 3 days).

6. Recommendations

1. **Plan bed capacity around scheduled admissions.** They stay more than twice as long as emergencies, so a small number of scheduled patients can use a large share of bed-days.
2. **Focus follow-up on chronic conditions in older patients.** Hypertension and diabetes patients aged 56 and over account for the longest stays, so early discharge planning and outpatient follow-up could shorten them.

3. **Review doctor workload.** Dr. Ade has the most patients and the longest average stay, so some cases could be shared with Dr. Bello or Dr. Musa.
4. **Collect more data before acting.** With 10 patients, these patterns are signals worth checking on a larger dataset, not firm conclusions.

7. Limitations

- The dataset has only **10 patients and 3 doctors**, so one patient can change an average noticeably.
- Admissions cover only **January 2026**, so seasonal patterns cannot be judged.
- Doctor comparisons depend on which patients each doctor happens to treat, so they say little about doctor performance.
- Treatment_Status was not part of the 25 queries, so the Recovered and Recovering pattern is an observation and not a tested result.

8. Conclusion

The queries show that length of stay is driven mainly by admission type, with scheduled admissions for chronic conditions in older patients staying the longest. Patients are spread evenly across conditions, and Dr. Ade carries the heaviest caseload. The analysis demonstrates filtering, aggregation, string and date functions, conditional logic and joins in SQL Server. The findings are based on a very small dataset, so they point to areas worth checking on more data and do not settle them.

Appendix: Query Summary

#	Task	Result
1	Patients from Lagos	3 patients
2	Length of stay over 5 days	3 patients
3	Age between 30 and 50	4 patients
4	Female, Lagos, Emergency	2 patients
5	Diabetes or Hypertension	4 patients
6	Top 3 longest stays	P007 (8), P004 (7), P009 (6)
7	Unique cities	8 cities
8	Total patients	10
9	Average age	44.6 years
10	Patients per condition	2 for each of 5 conditions
11	Names in uppercase	For example, AISHA BELLO
12	Names in lowercase	For example, aisha bello
13	Combine ID and name	For example, P001 Aisha Bello
14	Admitted after 15 Jan 2026	5 patients
15	Admission year and month	All January 2026
16	Average stay by admission type	Emergency 3.0, Scheduled 6.5
17	Average stay above 4 days	Scheduled only
18	Classify age groups	Young 4, Middle_age 4, Senior 2
19	Count per age group	Young 4, Middle_age 4, Senior 2
20	Short vs long stay	4 short, 6 long
21	Conditions with more than 1 patient	All 5 conditions
22	Patients with doctor and specialization	All 10 matched
23	Internal Medicine patients	3 patients (Dr. Musa)
24	Average stay per doctor	Ade 5.0, Bello 4.3, Musa 3.7
25	Doctor with the most patients	Dr. Ade (4 patients)